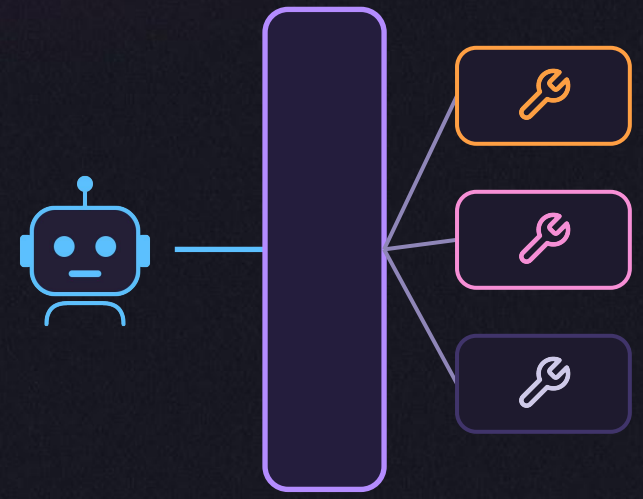
Temporal & AI Series

Agent Gateways with Temporal Nexus

One front door for every agent, with each team's tools plugged in over Nexus.



Dustin Sweet
Senior Solutions Architect, Temporal

Three things teams need from MCP tools

Every tool call an agent makes needs all three.



Governance

Only the tools you allow, with an audit log for every call.



Durability

Long tasks survive failure. Forward progress is never lost.



Human-in-the-loop

Pause for minutes or for weeks, until a person signs off.



What is an agent gateway?



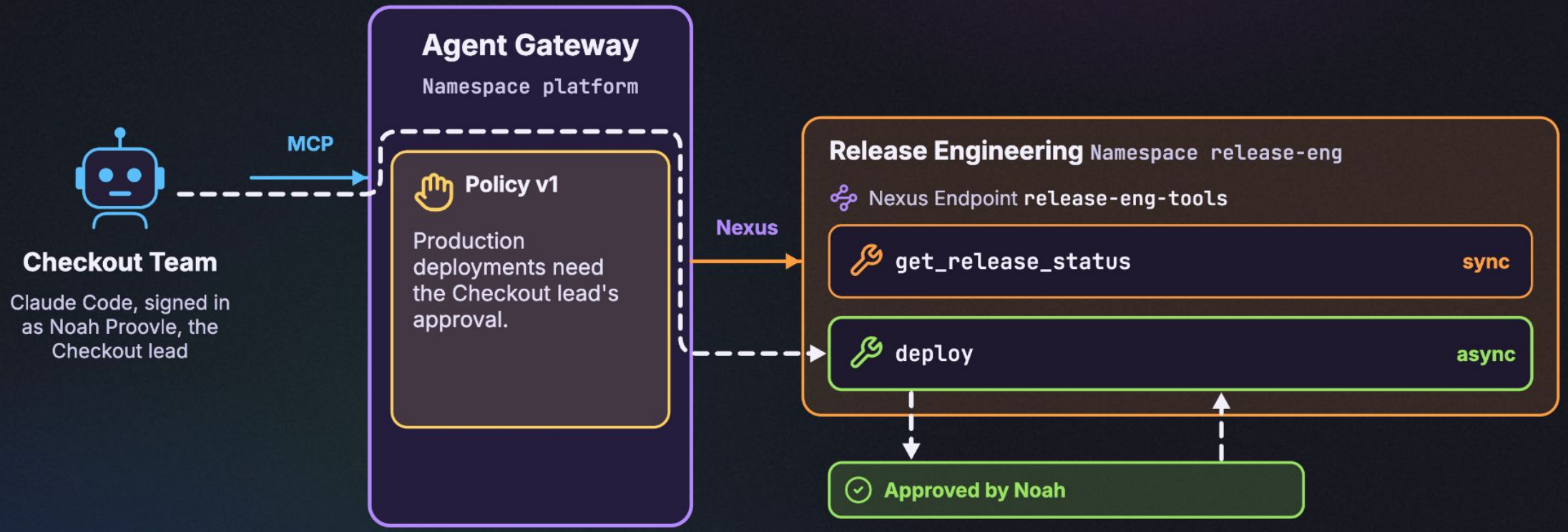
Teams plug in with Nexus

Each team owns its tools, in its own Namespace.



Our engineering team ships checkout-api through the gateway

Pinecrest Outfitters is a fictional outdoor retailer. Its Checkout team is our engineering team.



checkout-api staging 2.3.0 production 2.2.0 ready to ship: 2.3.1, 2.4.0

New mandate: no production deployment without Security sign-off

An unrelated incident elsewhere at Pinecrest just changed how every team ships to production.



Security team

Pinecrest's Security team doesn't own Checkout's service, and until now, they were never part of its release process.

Mandate: production releases

REQUIRED

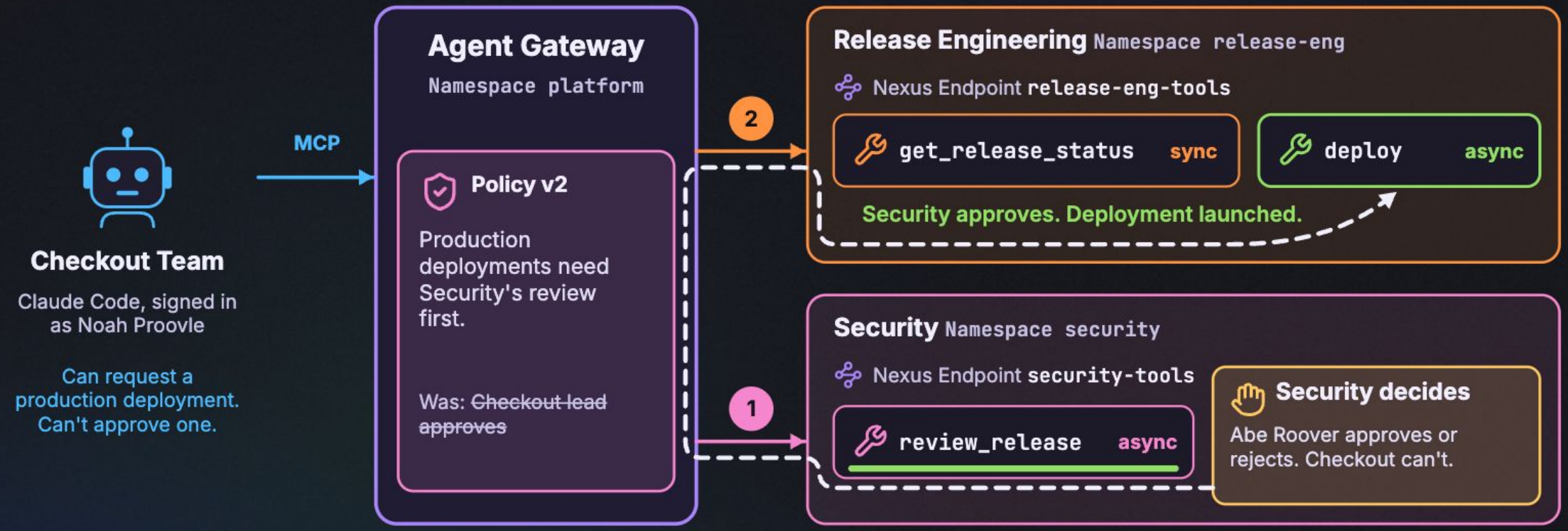
1. **No production deployment without a Security scan first.**
2. **Checkout no longer approves its own production releases.**
3. **Security approves the release, and initiates it themselves.**
4. **Checkout still owns and runs the release process end to end.**

Summary: only the final production gate changes hands, effective immediately.



After the mandate, Security decides what ships to production

Checkout's agent still asks. Security's review decides, in Security's own Namespace.



Summary: Checkout starts the review. Security owns the decision.

What to know before you build it

Six things worth knowing, and what a run costs.

- **No runtime discovery.** Nexus doesn't list a team's Operations, so the gateway keeps its own registry.
- **Namespaces, not people.** Nexus authorizes Namespaces. The gateway carries the person's identity; sign it in production.
- **A ten-second limit.** Synchronous Operations must finish in under 10 seconds. Longer work belongs in a Workflow.
- **Work can be retried.** Execution is at least once, so side effects need idempotency keys.
- **Allowlists are a Cloud feature.** Temporal Cloud enforces who can call each Endpoint. The local dev server doesn't.
- **Durable isn't correct.** Temporal keeps the reviewer's work safe, but a person still decides.

Cost to run. Under $0.50 for a full run with Claude Code and a real reviewer model. The key-free mode costs nothing.

The AI agent gateway owns the rules.
The teams own the tools.
Temporal Nexus is the contract between them.

Governance

Agents reach only the tools you allow, under policy, and every call is on the record.

Durability

Long-running work survives a crash and picks up exactly where it left off.

Human sign-off

Agents ask. The right person decides, and the request waits as long as it takes.

```
make setup  ·  make run  ·  make demo
```

github.com/temporal-community/temporal-ai-nexus-agent-gateway

Add your own tool: `docs/add-a-tool.md`

Thank you.